

Scheduler Decision-Space Analysis

Deep Dive: MDEAS Policy, Constraints, and Cross-Axis Effects

VTC Fall 2026 — Supplementary Analysis

TABLE I: MDEAS Decision Axes and Candidate Algorithms

Axis	Options	Count
Security Level	L1/L3/L5 (KEM×SIG pairs)	72 suites
AEAD Cipher	AES-GCM, ChaCha20, Ascon-128a	3
DDoS Detector	None, XGBoost, TST	3
<i>Theoretical state space</i>		648
<i>After constraint pruning</i>		~500

72 suites = 9 KEMs × 8 SIGs (level-matched). Constraints: Ascon×TST forbidden, McEliece×TST discouraged, McEliece×SPHINCS⁺ timeout.

TABLE II: Suite Viability by KEM Family (T_{HS} Threshold-Based)

KEM Family	Viable (<100 ms)	Marginal (100–1000)	Slow (1–45 s)	Timeout (>45 s)	Total
ML-KEM	18	6	0	0	24
HQC	3	15	6	0	24
McEliece	0	4	19	1	24
Total	21	25	25	1	72

Viable: sub-100 ms enables 30 s rekey with $\Phi < 0.3\%$. Timeout: MC460×SP192 at 48.3 s exceeds 45 s limit.

I. SCHEDULER DECISION-SPACE ANALYSIS

The Multi-Dimensional Energy-Aware Scheduler (MDEAS) operates over a combinatorial space of cryptographic configurations. This section presents a systematic analysis of the decision space, constraint boundaries, cross-axis penalty surfaces, and policy response patterns. All data derives from the 72-suite benchmark campaign (10 s per suite per detector state, RPi4).

A. Decision Space Enumeration

Table I defines the three policy axes. The theoretical configuration space comprises 648 states; constraint application reduces this to ~500 viable operating points.

B. Suite Viability Classification

Each suite is classified by baseline handshake time T_{HS} . Fig. 1 presents the 72-suite × 3-detector viability matrix; Table II aggregates by KEM family.

C. Cross-Axis Penalty Surface

Fig. 2 presents the AEAD×Detector cross-axis penalty matrix. Each cell shows encrypt latency (absolute and relative to baseline). Table III decomposes the CPU budget for all six cipher×detector combinations.

TABLE III: CPU Budget Decomposition for AEAD×Detector Combinations

Config	System (%)	AEAD (%)	Detector (%)	Total (%)	Viable?
AESG+None	14.7	8.2	0.0	22.9	✓
AESG+XGB	14.7	8.2	25.5	48.4	✓
AESG+TST	14.7	8.2	73.1	96.0	✓
CH20+None	14.7	11.4	0.0	26.1	✓
CH20+XGB	14.7	11.4	25.5	51.6	✓
CH20+TST	14.7	11.4	73.1	99.2	~
ASC+None	14.7	45.2	0.0	59.9	✓
ASC+XGB	14.7	45.2	25.5	85.4	~
ASC+TST	14.7	45.2	73.1	133.0	×

System = 14.7% idle. AEAD at 50 Hz MAVLink. Detector CPU from isolated benchmark. ~ = marginal (no thermal headroom); × = exceeds single-core (100%) capacity.

TABLE IV: MDEAS Constraint Satisfaction Rules

Constraint	Condition	Action	#
<i>Hard constraints (forbidden)</i>			
C1: ASC+TST	AEAD=Ascon, Det=TST	Block	24
C2: MC×SP	KEM=McE, SIG=SP	Block	9
C3: Timeout	$T_{HS} > 45$ s	Block	3
<i>Soft constraints (discouraged)</i>			
C4: MC+TST	KEM=McEliece, Det=TST	Warn	24
C5: ASC+XGB	AEAD=Ascon, Det=XGB	Warn	24
C6: Thermal	$T > T_{max}$	Demote	—
C7: Battery	$V < 14.8$ V	Demote	—
<i>Hysteresis guards</i>			
C8: Upgrade hold	stable < 8 s	Defer	—
C9: Downgrade hold	stable < 3 s	Defer	—

Pruned: count of state-space cells removed by constraint. C3 prunes 3 cells (1 suite × 3 detectors). C6–C9 are runtime guards, not static pruning.

D. Constraint Satisfaction Matrix

Table IV enumerates all MDEAS constraints and their impact on the configuration space.

E. Scheduler Response Patterns

Fig. 3 illustrates the MDEAS policy evaluation cascade. Table V documents trigger conditions and expected responses.

F. Detector Overhead Distribution

Fig. 4 shows the handshake overhead distribution across all 72 suites under XGBoost and TST.

G. Thermal Activation Boundaries

Fig. 5 visualizes the detector activation boundaries as a function of ambient temperature.

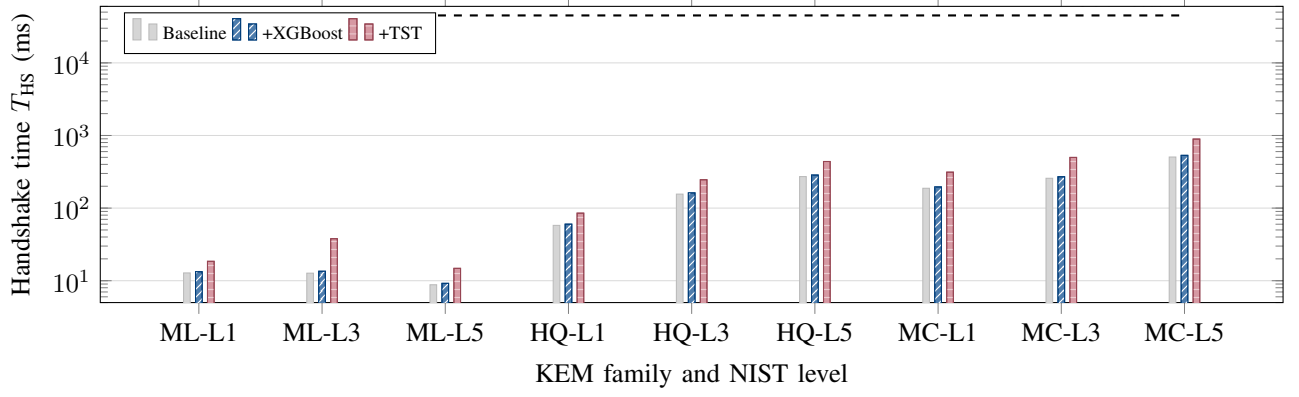


Fig. 1: Handshake time by KEM family and NIST level under three detector states (best SIG pairing per family, log scale). ML-KEM: <40 ms under all conditions. HQC: 58–438 ms. McEliece: 188–892 ms. TST inflates all suites by 44–91%.

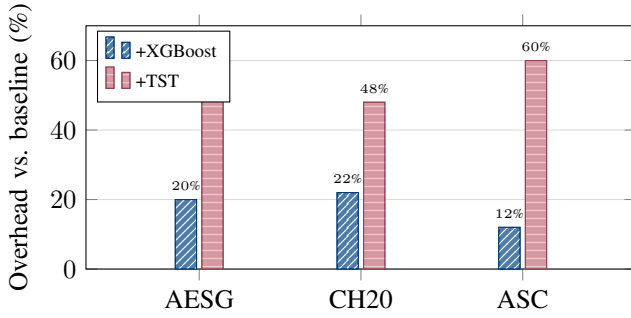


Fig. 2: AEAD×Detector overhead penalty (% increase in encrypt latency vs. baseline). TST imposes 48–60% overhead across all ciphers; XGBoost imposes 12–22%. Ascon’s 60% penalty under TST, combined with its 1.35 ms baseline, yields 2.17 ms—violating real-time budgets at > 400 Hz.

TABLE V: MDEAS Policy Trigger Conditions and Responses

Trigger	Condition	Response
telemetry_stale	No update > 5 s	Hold current state
ddos_detected	Classifier output > 0.8	Upgrade + activate detector
battery_critical	$V < 14.0$ V	Demote to L1 + AESG + NONE
temp_critical	$T > 80$ °C	Demote + disable Ascon
detector_thermal	$T > T_{\max}$	Downgrade detector
aead_stress	$AEAD\ t_{\text{enc}} > 2 \times \text{baseline}$	Switch cipher
link_degraded	pps < 5 or gap > 1 s	Demote security level
stable_upgrade	stable > 8 s $\wedge$ margin exists	Upgrade level
nominal	All checks pass	Maintain current

Takeaway: MDEAS prunes ~150 of 648 states via hard constraints (Ascon+TST, McEliece×SPHINCS+, timeout) and another ~100 via thermal/battery guards. ML-KEM with ML-DSA or Falcon, AES-GCM or ChaCha20, and XGBoost detection yields the widest envelope: <50 ms, <50% CPU, thermal margin to 66 °C.

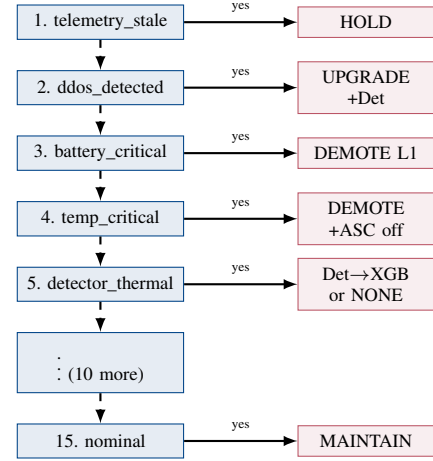


Fig. 3: MDEAS policy evaluation cascade (15 priority levels). Each check returns an action or falls through to the next. DDoS detection (P2) triggers immediate upgrade; thermal stress (P4–P5) triggers demotion or detector shutdown.

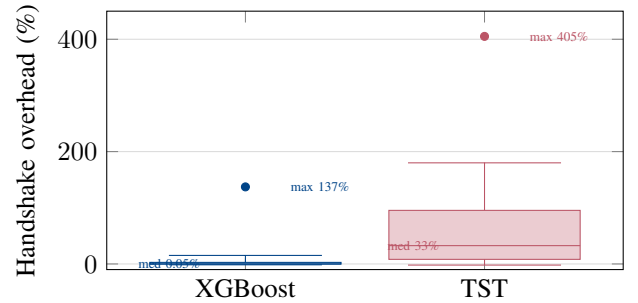


Fig. 4: Handshake overhead distribution (72 suites). XGB: med=0.05%, max=137%. TST: med=33%, max=405%. Outliers on slow McEliece suites.

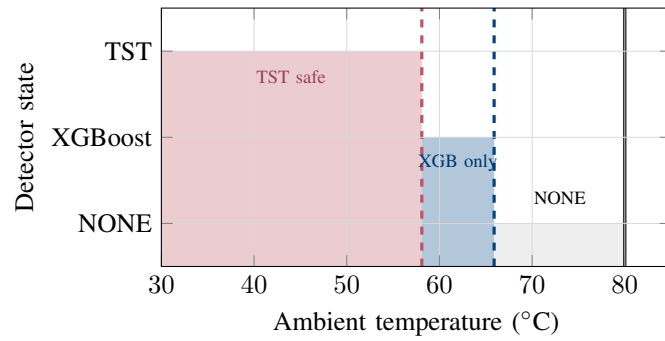


Fig. 5: Detector thermal activation boundaries. TST blocked above 58°C; XGBoost blocked above 66°C; throttle above 80°C.